

ZM-PBL Convection: CESM-CAM Case Configuration Summary

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Overview

This document summarises the eight CESM-CAM simulations used in the ZM-PBL convection study. All are AMIP-style (F-compset) integrations starting 1979-01-01 with 30-minute coupling (ATM_NCPL=48).

The experiment matrix probes three factors:

1. CAM6 vs CAM7 physics (compsets HIST_CAM60 vs HIST_CAM70%LT)
2. Vertical resolution in CAM7 (L32 / L48 / L58)
3. Zhang-McFarlane PBL-parcel source on (000a) vs off (001a)
controlled by namelist zmconv_parcel_pbl

Cases 1-7 share a single source tree (CAM_ZMKE, tag cam6_4_032-50, branch cam6_ke_zm); case 8 is the legacy CESM2.2 CAM6/FV baseline.

SourceMods: all cases contain only README files in SourceMods/src.cam and SourceMods/src.clm – no code overrides.

Table 1. Case family, grid, vertical resolution, tag

#	Case	Compset	Grid	Levels	CAM/CESM tag
1	f.e30.FLTHIST.CAM7.L32.000a	HIST_CAM70%LT	ne30pg3	32	cam6_4_032-50-gd4e4b939
2	f.e30.FLTHIST.CAM7.L48.000a	HIST_CAM70%LT	ne30pg3	48	cam6_4_032-50-gd4e4b939
3	f.e30.FLTHIST.CAM7.L58.000a	HIST_CAM70%LT	ne30pg3	58	cam6_4_032-50-gd4e4b939
4	f.e30.FLTHIST.CAM7.L32.001a	HIST_CAM70%LT	ne30pg3	32	cam6_4_032-50-gd4e4b939
5	f.e30.FLTHIST.CAM7.L48.001a	HIST_CAM70%LT	ne30pg3	48	cam6_4_032-50-gd4e4b939
6	f.e30.FLTHIST.CAM7.L58.001a	HIST_CAM70%LT	ne30pg3	58	cam6_4_032-50-gd4e4b939
7	f.e30.FHIST.CAM6-phys.L32.000a	HIST_CAM60	ne30pg3	32	cam6_4_032-50-gd4e4b939
8	f.e22.FHIST.f09_f09.CAM6.L32.000	HIST_CAM60 (+SIAC)	f09_f09 (FV 0.9x1.25)	32	cam_cesm2_2_rel_09-1-gc47c4d74

CAM7 compset resolves to HIST_CAM70%LT_CLM60%SP_CICE%PRES_DOCN%DOM_MOSART_SGLC_SWAV_SESP;
CAM6 compset resolves to HIST_CAM60_CLM50%SP_CICE%PRES_DOCN%DOM_MOSART_SGLC_SWAV_SESP (+SIAC for case 8).

Table 2. Physics options (differences from compset defaults)

Non-listed physics options are the compset default and identical across the group.

Cases	Deep conv	Shallow conv	Microphysics	ZM PBL parcel
1-3 CAM7 *.0000a (L32/L48/L58)	ZM	CLUBB_SGS	MG3 (+graupel)	on (default)
4-6 CAM7 *.001a (L32/L48/L58)	ZM	CLUBB_SGS	MG3 (+graupel)	OFF
7 CAM6-phys L32 (ne30, e30 tag)	ZM	CLUBB_SGS	MG2 (no graupel)	on
8 CAM6 f09 (CESM2.2)	ZM	CLUBB_SGS	MG2 (no graupel)	on

Table 3. user_nl_cam entries

#	Case	ncdata (initial conditions)	Physics tuning
1	L32.000a	FLT_L32_ne30pg3_IC_c220623.nc	(none)
2	L48.000a	FLT_L48_ne30pg3_IC_c220623.nc	(none)
3	L58.000a	(compset default)	(none)
4	L32.001a	FLT_L32_ne30pg3_IC_c220623.nc	zmconv_parcel_pbl = .false.
5	L48.001a	FLT_L48_ne30pg3_IC_c220623.nc	zmconv_parcel_pbl = .false.
6	L58.001a	(compset default)	zmconv_parcel_pbl = .false.
7	CAM6-phys L32	(compset default)	(none)
8	CAM6 f09	(compset default)	(none)

Common diagnostic tape settings (all cases):

```
mfilt = 0, 10, 60      nhtfrq = 0, -24, -3      ndens = 2, 2, 2
interpolate_output = .true. (cases 1-7 only; case 8 already on FV)
interpolate_nlat, interpolate_nlon = 192, 288
```

fincl history-tape variables include ZM convection diagnostics:

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ZMDT, ZMDQ, ZMMU, ZMMD, DTCOND, DCQ, CAPE, LCL/PLCL/TLCL/LEL, TKE, PCONVT, TROP_*, ...
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Case 8 (CESM2.2) omits LCL / PLCL / TLCL / LEL / BUOY / DMPDZ (variables only in newer CAM7 code).

Table 4. Run length

Case	STOP_N (years)	ATM_NCPL	Start date
L32.000a	4	48	1979-01-01
L48.000a	5	48	1979-01-01
L58.000a	2	48	1979-01-01
L32.001a	5	48	1979-01-01
L48.001a	5	48	1979-01-01
L58.001a	3	48	1979-01-01
CAM6-phys L32	10	48	1979-01-01
CAM6 f09	3	48	1979-01-01

Key takeaway

Cases 1-7 share a single source tree (CAM_ZMKE @ cam6_4_032-50, branch cam6_ke_zm). Differences are limited to:

- compset (CAM7 physics vs CAM6 physics)
- vertical grid (L32 / L48 / L58 for CAM7)
- one namelist knob zmconv_parcel_pbl (on for 000a, off for 001a)

This isolates the effect of the ZM deep-convection PBL-parcel source at three vertical resolutions, with CAM6 physics runs (cases 7-8) providing the pre-ZMKE baseline. Case 8 additionally tests the legacy CESM2.2 / finite-volume dycore combination against modern spectral-element ne30pg3.